Tiempo Rápido
(fc & fs & ts)

Tiempo Lento
(PRF & PRI)

Foto en t0
Foto en t0+PRI
Foto en t0+2PRI

100 x 20000

Read Signal

Raw Data

Matched Filter

Compressed
Signal

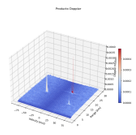
[m x m]

[n x m]

$(MTI_SC_Signal)^T$

$(Filtered_signal)^T$

FD(N) = exp(-j2π (row*col)/m)
row = col = [0, ..., m-1]



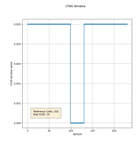
Filtrado Doppler

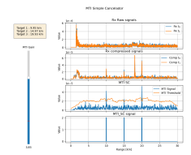
Target Velocity

MTI SC
Signal

MTI SC: H(z) = 1 - z^{-1}

CFAR Window
H_CFAR(t)





Detector

Target Range

MTI DC
Signal

MTI DC: H(z) = 1 - $2z^{-1}$ + z^{-2}

CFAR Window
H_CFAR(t)

Detector

Target Range

STI SC
Signal

STI SC: H(z) = 1 + z^{-1}

CFAR Window
H_CFAR(t)

Detector

Target Range